Harsh Sinha

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Education

International Institute of Information Technology Bangalore **August 2023 – July 2028**
Integrated Master of Technology in Computer Science and Engineering ***[CGPA - 3.64/4.00]***

Experience

Centre for Technology Research and Innovations for Digital Governance **May 2025 – July 2025**
Summer Intern *IIIT Bangalore*

- Extended **Spring Boot** backends for 2 applications (**Citizen Complaint Portal** & **Mess Management**) by engineering **10+ custom REST APIs** beyond **CRU**-only generated code, with filtered queries and multi-table joins.
- Secured APIs using **Spring Security** by integrating **Google OAuth 2.0** for authentication, issuing custom **JWT**s post-login, and enforcing **RBAC** across **4 roles** via a JWT filter chain.

Projects

Food-e | *Node.js (Express), MongoDB, Redis, Cloudinary, Microsoft OAuth* **August 2025 – Present**

- Engineered the **complete backend** for a **college-wide mess management platform**, officially inaugurated by the **Institute Director** on **Independence Day 2025**, serving **1,600+ registered users** with ∼**900 daily active users** and a **61% retention rate**.
- Designed **role-based functionality** for **4 roles** (Users, Admins, Committee Members, Caterer Managers), enabling ratings, image feedback, sick-meal requests, announcements, and **PDF report generation** with per-item avg ratings.
- Implemented secure auth using **JWT** (access & refresh tokens) and **Microsoft OAuth**, restricting access to verified college users, with **425+ feedback submissions** processed to date.
- Optimised performance using **Redis caching**, **Cloudinary** for file storage, **cron jobs** for automated cleanup, and **MVC architecture** for maintainability.

AcadOS | *C, System Calls, File Locking, Linux* **May 2025 – June 2025**

- Developed a concurrent **TCP client-server** system in **C** using **POSIX pthreads** with a **thread-per-connection** model, sustaining **26,000+ RPS** under **500 concurrent clients** with sub-**6ms** median latency.
- Engineered **file-based persistence** with **POSIX semaphores** and **flock()** for atomic writes, **role-based authentication** (Admin, Faculty, Student) with **termios** password masking, and **SIGINT** handling to flush sessions.
- Stress-tested across **11 concurrency levels** (5–500 clients, **15,000 transactions**), achieving peak **32,572 RPS** and sub-**0.34ms** p99 tail latency at low concurrency.

EcoSynergy | *Node.js, Express, MongoDB, JWT, AI Integration* **April 2025 – June 2025**

- Designed **backend APIs** for a **MERN platform** enabling creation, joining, and management of community sustainability initiatives with **role-based access** and **JWT + Passport.js** authentication.
- Integrated **Socket.IO** for real-time event broadcasting (drive creation, updates, cancellations, and live chat) with **Redis pub/sub** for message caching and resilience.
- Enforced security using **Helmet, rate-limiting, mongo-sanitize**, and **Joi** validation, with centralized error handling and **Gemini API** for AI-powered post-event report summarization.

Achievements

Dean's Merit List | *IIIT Bangalore* **June 2024 – Present**

- Awarded by the Dean of Academics for maintaining a **CGPA of 3.6+/4.0 (9/10)** for **2 consecutive years** (2024 & 2025).

Mentorship & Leadership

Teaching Assistant **August 2025 – May 2026**
C Programming & Computer Networks *IIIT Bangalore*

∗ Assisted **370+ students** over **40+ sessions** in **C Programming** & **Computer Networks**, conducting labs and grading assessments.

Mentor **August 2025 – Present**
Food Committee *IIIT Bangalore*

∗ Mentored **3 juniors** to independently ship the **announcement feature** on **Food-E**.

Technical Skills

Languages: Java, Python, C, C++, MySql, JavaScript, OCaml
Technologies/Frameworks: SpringBoot, Node.js, Express.js, Next.js, ReactJS
Databases: PostgreSQL, SQL, MongoDB, Redis